



NORTH SYDNEY BOYS HIGH SCHOOL

2010 HSC ASSESSMENT TASK 2

Mathematics Extension 1

General Instructions

- Working time – 60 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Ireland
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Barrett
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	Total	Total %
Mark	$\frac{\quad}{17}$	$\frac{\quad}{13}$	$\frac{\quad}{12}$	$\frac{\quad}{10}$	$\frac{\quad}{14}$	$\frac{\quad}{66}$	$\frac{\quad}{100}$

Question 1 (17 marks)**Marks**

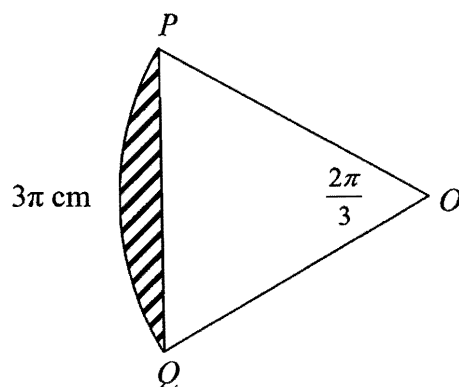
- (a) Evaluate $\frac{\pi}{e^3}$ correct to 3 significant figures. 1
- (b) Find $\frac{dy}{dx}$ for the following:
- (i) $y = x^2 e^{3x}$ 2
- (ii) $y = \sqrt{\ln 2x}$ 2
- (iii) $y = 3^x$ 1
- (c) (i) Find $\int \frac{x^2}{x^3 + 3} dx$ 2
- (ii) Evaluate $\int_0^1 x^2 e^{3x} dx$ 2
- (d) If $y = \ln\left(\frac{e^x + 1}{e^x - 1}\right)$, find $\frac{dy}{dx}$ as a fraction in simplest form. 3
- (e) If $f(x) = \ln(x - 3)(5 - x)$, what is the domain of $f(x)$? 2
- (f) Solve the equation $2 \ln(x - 1) = \ln(4x - 7)$. 2

Question 2 (13 marks)

- (a) Use Mathematical Induction to prove that $4^n - 1$ is divisible by 3 for all $n \geq 1$. 5
- (b) (i) Show that the equation $x^3 + 2x - 4 = 0$ has a root α in the interval $1 < \alpha < 2$. 1
- (ii) Show that this equation has **exactly** one root in this interval. (Hint: You might want to consider the derivative.) 2
- (iii) Use Newton's method to show that if $x_1 = a$ is an approximation to α , then the next approximation is $x_2 = \frac{2a^3 + 4}{3a^2 + 2}$. 3
- (iv) Use Newton's method once with an initial approximation of $x_1 = 2$ to find a better approximation for α . (Give your answer correct to 2 decimal places.) 2

Question 3 (12 marks)

(a)



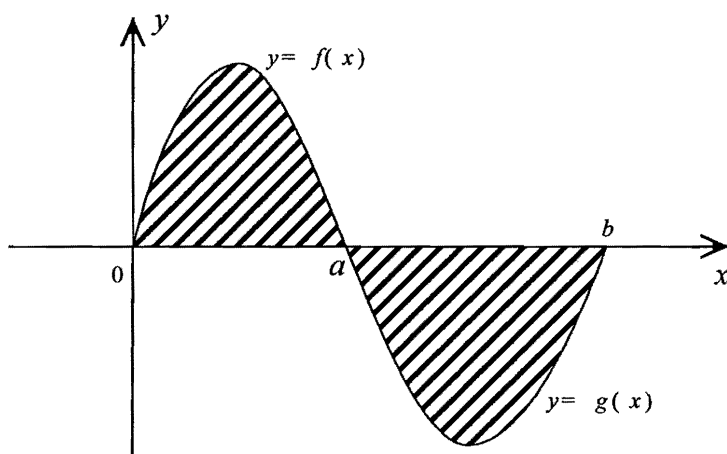
In the diagram above, PQ is the arc of a circle, centre O .

The length of the arc is 3π cm, and angle POQ is $\frac{2\pi}{3}$.

- | | | |
|-------|--|---|
| (i) | Find the radius of the circle. | 2 |
| (ii) | Find the area of the sector OPQ . | 2 |
| (iii) | Find the area of the shaded region. | 2 |
| (b) | (i) State the period and amplitude of the function $y = 3 - \cos 2x$. | 2 |
| | (ii) Sketch the curve $y = 3 - \cos 2x$ for $0 \leq x \leq 2\pi$. | 3 |
| (c) | Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$. | 1 |

Question 4 (10 marks)

(a)



Write an expression involving integrals for the area of the shaded region.

(b) Find the area enclosed between the hyperbola $y = \frac{4}{x}$ and the line $y = 5 - x$. 4

(c) (i) Differentiate $y = \ln(x^2)$. 1

(ii) Hence evaluate $\int_e^{e^2} \frac{dx}{x \ln(x^2)}$. 3

Question 5 (14 marks)

(a) (i) Find the stationary point on the curve $y = e^x + xe^x$, and determine its nature. 3

(ii) Find any points of inflexion. 3

(iii) Find where the above curve cuts the coordinate axes. 2

(iv) Sketch the curve $y = e^x + xe^x$, . 2

(b) Find the volume of the solid generated when the region bounded by the curve $y = \ln x$, the x -axis, and the lines $x = 1$ and $x = 2$ is rotated about the y -axis. 4



Q1) a) 0.156

e) $f(x) = \ln(x-3)(5-x)$
 $(x-3)(5-x) > 0$

b) i) $y = x^2 e^{x^3}$
 $\frac{dy}{dx} = 2x e^{x^3} + x^2 \cdot 3x^2 e^{x^3}$
 $\frac{dy}{dx} = x e^{x^3} (2 + 3x^3)$

D: $3 < x < 5$

f) $\ln(x-1)^2 = \ln(4x-7)$
 $(x-1)^2 = 4x-7$
 $x^2 - 2x + 1 = 4x - 7$
 $x^2 - 6x + 8 = 0$
 $(x-4)(x-2) = 0$
 $\therefore x = 2 \text{ or } 4$

ii) $y = (\ln 2x)^{1/2}$
 $\frac{dy}{dx} = \frac{1}{2} (\ln 2x)^{-1/2} \cdot \frac{2}{2x}$
 $= \frac{1}{2x(\ln 2x)^{1/2}}$

iii) $y = 3^x$
 $\ln y = x \ln 3$
 $\frac{1}{y} \frac{dy}{dx} = \ln 3$
 $\frac{dy}{dx} = \ln 3 \cdot 3^x$

c) i) $\int \frac{3x^2}{x^3+3} dx$
 $= \frac{1}{3} \ln(x^3+3) + C$

ii) $\int_0^1 3x^2 e^{x^3} dx$
 $= \frac{1}{3} e^{x^3} \Big|_0^1$
 $= \frac{1}{3} e - \frac{1}{3}$

d) $y = \ln(e^x+1) - \ln(e^x-1)$
 $\frac{dy}{dx} = \frac{e^x}{e^x+1} - \frac{e^x}{e^x-1}$
 $= \frac{e^{2x} - e^2 - e^{2x} - e^x}{(e^x+1)(e^x-1)}$
 $= \frac{-2e^x}{(e^x+1)(e^x-1)}$

Question 2 a) Two methods shown:

When $n=1$:

$$4^1 - 1 = 3$$

∴ True when $n=1$ ✓

Assume true for $n=k$

$$4^k - 1 = 3N \text{ where } N \text{ is an integer}$$

Prove true for $n=k+1$

$$4^{k+1} - 1 = 4 \cdot 4^k - 1$$

$$= 4(4^k - 1) + 3$$

$$= 4(3N) + 3$$

$$= 3(4N + 1)$$

∴ True for $n=k+1$ if true for $n=k$ ✓

Since true for $n=1$, also true for $n=2$, and also for $n=3$ and so on. ✓

∴ $4^n - 1$ is divisible by 3 for all $n \geq 1$.

b) i/ $x^3 + 2x - 4 = 0$

let $f(x) = x^3 + 2x - 4$

$$f(1) = -1 < 0$$

$$f(2) = 8 > 0$$

∴ root between $x=1$ and 2

ii/ $f'(x) = 3x^2 + 2$

> 0 for all x ✓

∴ curve ~~increases up for all~~ is increasing for all x ✓

∴ crosses x -axis once

(Alt: show no solution to $3x^2 + 2 = 0$; then state this means no stationary/turning points).

Q 2

a) Let $SC(n)$ be statement that $4^n - 1$ is divisible by 3 for $n \geq 1$

For $n=1$:

$$LHS = 4^1 - 1 = 3$$

i.e $SC(1)$ is true ✓

Assume $SC(k)$ is true

i.e $4^k - 1 = 3M$ where M is an integer

$$\therefore 4^k = 3M + 1$$

RTP $SC(k+1)$ is true

i.e $4^{k+1} - 1$ is divisible by 3 ✓

$$4^{k+1} - 1 = 4 \cdot 4^k - 1$$

$$= 4 \cdot 4^k - 1$$

$$= 4(3M + 1) - 1$$

$$= 12M + 3$$

$$= 3(4M + 1) \text{ which is divisible by 3 since } 4M + 1 \text{ is an integer}$$

∴ If $SC(k)$ is true, then $SC(k+1)$ is true ✓

Since $SC(1)$ is true, and then $SC(2)$ is true, and $SC(3)$ is true and so on

∴ By M.I., $SC(n)$ is true for all $n \geq 1$ ✓

$$x_2 = a - \frac{a^3 + 2a - 4}{3a^2 + 2} \quad \checkmark$$

$$= \frac{3a^3 + 2a - a^3 - 2a + 4}{3a^2 + 2}$$

$$= \frac{2a^3 + 4}{3a^2 + 2} \quad \checkmark$$

$$x_2 = \frac{2(2)^3 + 4}{3(2)^2 + 2} \quad \checkmark$$

$$= 1.43 \text{ to } 2 \quad \checkmark$$



Q3)

a) i/ $\ell = r\theta$

$$3\pi = r \frac{2\pi}{3}$$

$$\therefore r = \frac{3\pi \times 3}{2\pi}$$

$$= \frac{9}{2} \text{ cm}$$

ii/ $A = \frac{1}{2} r^2 \theta$

$$= \frac{1}{2} \times \frac{81}{4} \times \frac{2\pi}{3}$$

$$= \frac{27\pi}{4} \text{ cm}^2 \text{ or } 21.21$$

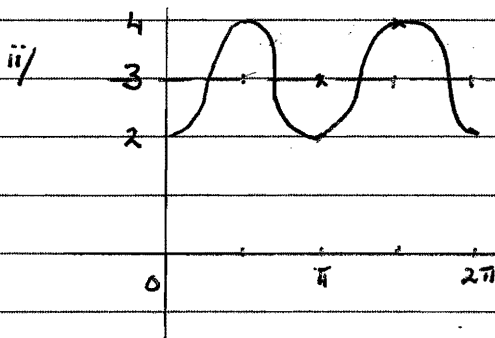
iii/ $A = \frac{27\pi}{4} - \frac{1}{2} \times \frac{9}{2} \times \frac{9}{2} \frac{\sin 2\pi}{3}$

$$= \frac{27\pi}{4} - \frac{81}{8} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{27\pi}{4} - \frac{81\sqrt{3}}{16} \text{ cm}^2 \text{ or } \frac{108\pi - 81\sqrt{3}}{16} \text{ or } 12.437$$

b) i/ $\text{amp} = 1$

$$\text{period} = \pi$$



c) $2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$

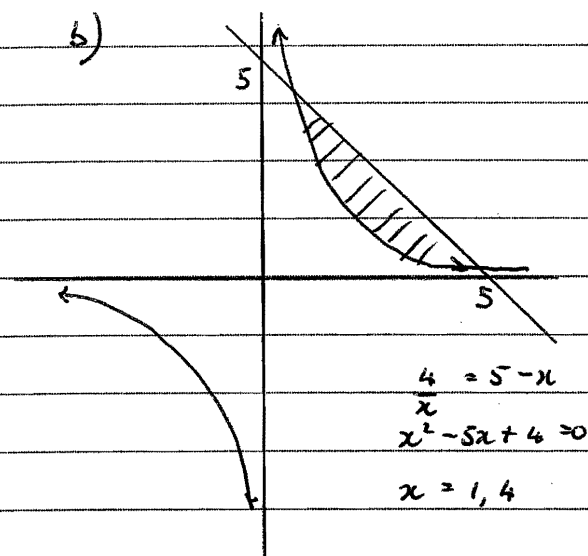
$$= 2 \times 1$$

$$= 2$$



Q4) a) $A = \int_0^a f(x) dx + \left| \int_a^5 g(x) dx \right|$ or $A = \int_0^a f(x) dx - \int_a^5 g(x) dx$

b)



$$A = \int_1^4 \left(5 - x - \frac{4}{x} \right) dx$$

$$= \left[5x - \frac{x^2}{2} - 4 \ln x \right]_1^4$$

$$= 20 - 8 - 4 \ln 4 - 5 + \frac{1}{2}$$

$$= \frac{15 - 4 \ln 4}{2}$$

c) i/ $y = \ln(x^2)$

$$\frac{dy}{dx} = \frac{2}{x}$$

ii/ $\frac{1}{2} \int_e^{e^2} \frac{\frac{2}{x}}{(\ln x)^2} dx$

$$= \frac{1}{2} \ln \left[\ln x^2 \right]_e^{e^2} = \frac{1}{2} \left[\ln(\ln e^4) - \ln(\ln e^2) \right] = \frac{1}{2} (\ln 4 - \ln 2)$$

$$= \frac{1}{2} \ln 2$$

The hence has to be used.

Students who work by substitution and get the correct answer, lose 1 or 2 marks.



Q5)

a)

i/

$$y = e^x + xe^x$$

$$\frac{dy}{dx} = e^x + e^x + xe^x$$

$$= 2e^x + xe^x$$

$$= e^x(2+x)$$

$$\frac{dy}{dx} = 0 \text{ when } x = -2$$

$$\frac{d^2y}{dx^2} = 2e^x + e^x + xe^x$$

$$= 3e^x + xe^x$$

$$\text{when } x = -2$$

$$\frac{d^2y}{dx^2} = \frac{3}{e^2} - \frac{2}{e^2} = \frac{1}{e^2}$$

$$> 0$$

$$\therefore \text{min at } (-2, \frac{-1}{e^2})$$

ii/

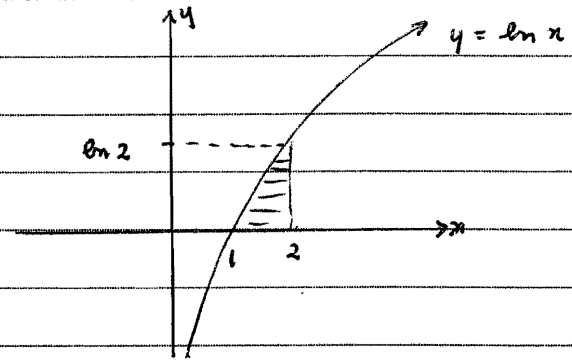
$$\frac{d^2y}{dx^2} = 0 \text{ when } x = -3$$

$$x = -4 - 3 - 2$$

$$\frac{d^2y}{dx^2} < 0 = 0 > 0$$

$$\therefore \text{inf at } (-3, -\frac{2}{e^3})$$

b)



$$V = \pi \int x^2 dy$$

$$= 4\pi \ln 2 - \pi \int_0^{\ln 2} e^{2y} dy$$

$$= 4\pi \ln 2 - \pi \left[\frac{1}{2} e^{2y} \right]_0^{\ln 2}$$

$$= 4\pi \ln 2 - \frac{\pi}{2} [4 - 1]$$

$$= \left(4\pi \ln 2 - \frac{3\pi}{2} \right) u^3$$

